EULER CHARACTERISTIC DUALITY FOR PROPER CURVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument:

Editorial note. The label lemma-euler names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The dualizing complex and module of a curve. Same chapter, \label{lemma-duality-dim-1}. Statement lines 1176-1178: omega_X^bullet and omega_X are as in this lemma; the right hand side chi(X, omega_X^bullet) is defined via them.
- The dualizing complex in the Cohen-Macaulay case. Same chapter, \label{lemma-duality-dim-1-CM}. Proof lines 1200-1203: in the CM equidimensional case omega_X^bullet = omega_X[1], which yields the second formula from the first.

Source reference. Original reference 0BS2: sources/stacks-curves/pinned-original-source.tex, source lines 796–796. Prerequisite pointer invoked at proof line(s) 1176; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS3: sources/stacks-curves/pinned-original-source.tex, source lines 864–864. Prerequisite pointer invoked at proof line(s) 1203; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a proper scheme of dimension ≤ 1 over a field k. With $\omega_X^\bullet$ and ω_X as in Lemma 0BS2 we have*

$$\chi(X, \mathcal{O}_X) = \chi(X, \omega_X^\bullet)$$

If X is Cohen-Macaulay and equidimensional of dimension 1, then

$$\chi(X, \mathcal{O}_X) = -\chi(X, \omega_X)$$

Proof. We define the right hand side of the first formula as follows:

$$\chi(X, \omega_X^\bullet) = \sum\nolimits_{i \in \mathbf{Z}} (-1)^i \dim_k H^i(X, \omega_X^\bullet)$$

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This is well defined because $\omega_X^\bullet$ is in $D^b_{Coh}(\mathcal{O}_X)$, but also because

$$H^i(X, \omega_X^\bullet) = \mathrm{Ext}^i(\mathcal{O}_X, \omega_X^\bullet) = H^{-i}(X, \mathcal{O}_X)$$

which is always finite dimensional and nonzero only if $i = 0, -1$. This of course also proves the first formula. The second is a consequence of the first because $\omega_X^\bullet = \omega_X[1]$ in the CM case, see Lemma 0BS3. $\square$